

MANILA NINOY AQUINO INTL, Philippines

WMO: 984290

Lat: 14.509N

Lon: 121.020E

Elev: 75

StdP: 14.66

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 71.3	(c) 73.0	(d) 62.0	(e) 83.4	(f) 80.5	(g) 63.1	(h) 86.7	(i) 80.2	(j) 18.7	(k) 83.7	(l) 16.5	(m) 83.8	(n) 2.7	(o) 70	(p) 0.385

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 5	(b) 13.1	(c) 95.2	(d) 78.6	(e) 93.8	(f) 78.5	(g) 92.9	(h) 78.4	(i) 83.3	(j) 87.0	(k) 82.6	(l) 86.4	(m) 82.0	(n) 86.1	(o) 9.4	(p) 120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
82.6	170.6	84.6	82.1	167.3	84.3	80.9	161.1	83.8	47.4	87.5	46.7	86.3	46.1	86.2	88.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 18.9	(b) 16.5	(c) 14.4	(e) 68.6	(f) 98.2	(g) 2.0	(h) 1.5	(i) 67.1	(j) 99.2	(k) 66.0	(l) 100.1	(m) 64.9	(n) 100.9	(o) 63.4	(p) 102.0		
(4) 18.9	(5) 16.5	(4) 14.4	(4) 68.6	(4) 98.2	(4) 2.0	(4) 1.5	(4) 67.1	(4) 99.2	(4) 66.0	(4) 100.1	(4) 64.9	(4) 100.9	(4) 63.4	(4) 102.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	82.8	80.0	81.0	83.1	85.9	86.2	84.5	82.8	82.3	82.4	82.6	82.3	80.9
	DBStd	2.83	2.03	2.09	2.13	2.13	2.60	2.43	2.33	2.03	1.93	1.85	1.95	2.25
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11986	930	867	1027	1078	1122	1035	1015	1002	971	1012	970	957
	CDD65	6511	465	447	562	628	657	585	550	537	521	547	520	492
	CDH74	72768	4193	4376	6409	8206	8564	7134	6035	5846	5595	6069	5604	4737
	CDH80	27709	1140	1406	2548	4053	4252	3081	2120	1917	1814	2102	1917	1359
Wind	WSAvg	6.1	5.8	6.6	7.1	7.2	6.5	6.0	6.2	6.9	6.0	5.1	5.0	5.2
Precipitation	PrecAvg	86.6	1.0	0.9	1.1	1.0	5.8	8.7	16.3	18.2	15.3	9.0	4.8	4.3
	PrecMax	133.0	4.0	5.0	4.8	6.3	16.6	21.9	48.6	43.8	35.4	18.7	10.9	15.6
	PrecMin	56.8	0.0	0.0	0.0	0.0	0.3	2.3	4.2	9.0	6.1	2.1	0.0	0.0
	PrecStd	18.9	1.1	1.3	1.3	1.4	3.9	4.4	9.2	8.4	5.7	4.0	3.1	3.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.7	91.6	93.6	96.9	97.2	95.4	93.3	91.6	91.7	91.8	91.7	91.2
		MCWB	76.2	76.2	77.2	77.7	78.5	80.1	79.7	80.3	80.0	78.9	78.1	77.0
	2%	DB	88.0	89.8	92.1	95.1	95.4	93.4	91.3	89.7	89.8	90.4	90.0	89.4
		MCWB	75.8	75.9	76.9	77.8	78.8	80.1	79.8	79.9	80.1	78.8	78.0	76.9
	5%	DB	86.6	88.3	91.2	93.6	94.3	91.6	89.6	88.1	88.2	89.2	89.4	87.7
		MCWB	75.3	75.4	76.6	77.5	78.8	79.9	79.7	80.0	80.3	78.7	78.0	76.8
	10%	DB	85.7	87.3	89.6	92.7	92.8	89.9	87.9	87.4	87.4	87.7	87.8	86.1
		MCWB	74.9	75.1	76.0	77.4	79.0	79.8	79.8	79.9	80.2	78.7	78.0	76.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.6	79.8	80.4	82.5	83.6	84.1	83.9	83.6	83.8	83.5	82.5	81.0
		MCDB	84.0	85.6	87.8	87.5	88.7	88.3	86.6	86.6	86.7	86.7	86.3	84.1
	2%	WB	78.5	78.4	79.2	80.9	82.3	82.7	82.6	82.6	82.7	82.3	80.9	79.8
		MCDB	83.5	84.7	87.0	88.1	87.8	87.1	86.1	85.7	85.9	85.3	85.4	83.7
	5%	WB	77.5	77.5	78.5	80.0	81.5	81.9	81.7	81.7	81.9	81.3	80.1	79.0
		MCDB	82.9	84.3	86.5	88.1	87.7	86.7	85.7	85.2	85.3	84.8	84.8	83.2
	10%	WB	76.6	76.5	77.7	79.2	80.7	81.1	80.8	80.8	81.0	80.6	79.2	78.2
		MCDB	82.3	83.7	85.7	87.9	87.6	86.5	85.3	84.7	84.6	84.3	83.9	82.8
Mean Daily Temperature Range	5% DB	MDBR	12.2	13.1	14.0	14.4	13.1	11.7	10.5	9.8	10.1	10.9	11.6	11.2
		MCDBR	13.3	14.5	15.2	15.4	14.9	13.4	12.7	11.3	11.6	12.8	13.4	12.9
		MCWBR	5.5	5.6	5.7	5.5	5.2	5.1	5.3	5.2	5.4	5.4	5.3	5.2
		MCDDBR	11.7	12.8	13.8	14.1	13.0	11.9	10.9	10.1	10.5	10.7	11.6	10.9
	5% WB	MCWBR	5.9	5.8	5.9	5.8	5.5	5.4	5.8	5.7	5.7	5.7	5.6	5.5
Clear-Sky Solar Irradiance	taub	0.421	0.419	0.450	0.474	0.474	0.467	0.463	0.470	0.466	0.477	0.439	0.441	
	taud	2.416	2.409	2.325	2.262	2.292	2.327	2.339	2.308	2.316	2.266	2.387	2.367	
	Ebn at Noon	277	283	276	267	263	262	264	265	267	262	269	267	
	Edh at Noon	36	38	42	45	43	41	40	42	42	43	37	37	
All-Sky Solar Radiation	RadAvg	1331	1577	1782	2006	1813	1627	1433	1387	1467	1408	1321	1183	
	RadStd	82	138	124	108	162	161	182	142	147	115	117	104	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	+0.76	+1.48	N/A	+1.04	N/A	N/A	N/A	N/A	+329	+329
Regional (16 neighbors)	+0.58	+0.85	N/A	+0.81	+0.72	+0.97	N/A	N/A	+284	+284

Nomenclature: See separate page